Tian Lan （兰天）

Curriculum Vitae — Prepared for IEEE Senior Member Application

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IEEE Membership No.: ____________________ (fill in) | Candidate field: Computer Science / Artificial Intelligence

This document follows the structure recommended for IEEE Senior Member elevation packages: (1) education and employment listed with month/year and duties, (2) an explicit tally of the ten-year professional-practice requirement (IEEE Bylaw I-104.3), and (3) a dedicated "Significant Performance" section with discrete, dated project write-ups covering at least 60 months. All dates below should be independently verified before submission to nominate.ieee.org.

Before submitting, please double-check these dates (source documents disagree slightly):
IBM Research China months (shown as Jun.–Aug. 2019, not otherwise documented) · Tencent AI Lab end date (Jan. 2023 here vs. Mar. 2023 in an earlier resume) · miHoYo dates (Feb.–Jun. 2023 here vs. Mar.–Jul. 2023 in an earlier resume) · Shanghai AI Lab end date (Jan. 2025 here vs. Jul. 2024 in the 2026 visa CV).

Profile

I am currently an Algorithm Expert at Alibaba Group (Token Hub · MaaS), focusing on automatic evaluation and long-horizon agent technologies for large language models, spanning both frontier research and real-world business applications. I received my Ph.D. in Computer Science and Technology from Beijing Institute of Technology, advised by Prof. Xian-Ling Mao, and have published 20+ peer-reviewed papers at venues including ACL, EMNLP, NeurIPS, ICLR, ICML, and SIGKDD, accumulating 1,700+ citations.

Summary for the Admission & Advancement Review Panel

Field: Computer Science and Information Technology (Natural Language Processing / Machine Learning), an IEEE-designated field.
Ten years of professional practice: B.Eng. (Beijing Institute of Technology, Sep. 2015 – Jun. 2019) credited as **3 years**, plus continuous professional research practice (industrial research internships and Ph.D.-level research with a sustained publication record) from **Jul. 2019 to present** (∼7 years as of Aug. 2026) ⇒ **total ≈ 10 years and growing**.
Five years (60+ months) of significant performance: Jun. 2021 – present (∼62 months), documented in the "Significant Performance" section below via three discrete, dated projects.
References: to be provided by three current IEEE Senior Members / Fellows / Honorary Members familiar with the candidate's work.

Education

Beijing Institute of Technology — Beijing, China
Ph.D. in Computer Science and Technology (Combined Master–Ph.D. Program) *Sep. 2019 – Jun. 2025*
Advisor: Prof. Xian-Ling Mao. Dissertation area: automatic evaluation of text generation and LLMs. Successfully defended dissertation on Jun. 20, 2025.

Beijing Institute of Technology — Beijing, China
B.Eng. in Computer Science and Technology *Sep. 2015 – Jun. 2019*
(Credited as 3 years toward the ten-year professional-practice requirement.)

Professional Experience (Employment History)

Alibaba Group — Hangzhou, China (Full-time employment)
Algorithm Expert, Token Hub (ATH) – MaaS, Agent Systems *Aug. 2025 – Present*
- Lead researcher for agentic-AI evaluation and long-horizon agent research within Alibaba's Marco DeepResearch initiative.
- Design large-scale agent benchmarks (e.g., HSCodeComp, Table-as-Search) and direct their end-to-end research, from problem formulation and expert-panel data annotation to publication and open-source release.
- Coordinate a cross-functional team including annotators, engineers, and co-authors across research and product organizations.

Tencent TEG — Shenzhen, China (Internship)

Research Intern, AI Platform Department *Apr. 2025 – Jun. 2025*

- Conducted research on AI applications in gaming and contributed to the Block-Attention efficient-prefilling method (ICLR 2025).

Shanghai Artificial Intelligence Laboratory — Shanghai, China (Internship)

Research Intern, InternLM Post-training Group *Jul. 2023 – Jan. 2025*

- Designed and led **CriticEval**, a systematic benchmark for evaluating large language models' critique ability (four dimensions, nine scenarios); results published at NeurIPS 2024.
- Proposed a multi-agent feedback framework for training language models to critique, adopted in the post-training pipeline of the open-sourced InternLM model family; published at EMNLP 2025 (Findings).

miHoYo — Shanghai, China (Internship)

Research Intern, Natural Language Processing Lab *Feb. 2023 – Jun. 2023*

- Developed AI applications for gaming environments and domain-specific model post-training pipelines.

Tencent AI Lab — Shenzhen, China (Internship)

Research Intern *Jun. 2021 – Jan. 2023*

- Designed the Copy-Generator architecture (*Copy is All You Need*), a copy-based neural text-generation paradigm; published at ICLR 2023.
- Co-developed the Contrastive Search / SimCTG decoding method (NeurIPS 2022, Spotlight); the method was subsequently adopted as a built-in decoding strategy in HuggingFace's `transformers` library (v4.24.0), a widely used open-source machine-learning framework.
- Contributed to Effidit, Tencent AI Lab's AI-powered writing-assistant product.

IBM Research – China — Beijing, China (Internship)

Research Intern *Jun. 2019 – Aug. 2019*

- Conducted research on dialogue systems.

Note: all positions prior to Alibaba Group (Tencent AI Lab, Tencent TEG, Shanghai AI Lab, miHoYo, IBM Research China) were research internships undertaken alongside full-time Ph.D. study; Alibaba Group (Aug. 2025 – present) is my first full-time employment. Sep. 2019 – Jun. 2025 overlaps with full-time Ph.D. study at Beijing Institute of Technology. Per IEEE guidance, this period is counted once, as continuous professional research practice (not additionally claimed as doctoral-degree credit), consistent with sustained industrial research internships and an uninterrupted, published research record throughout.

Significant Performance (60+ Months)

1. Efficient and copy-augmented text generation — Tencent AI Lab *Jun. 2021 – Jan. 2023 (20 months)*
Task: Improve the efficiency and factual grounding of neural text generation. **My role:** Sole technical lead; designed the Copy-Generator retrieval-augmented decoding architecture and co-designed the Contrastive Search decoding algorithm. **Why significant:** Copy-Generator was published at ICLR 2023 (180+ GitHub stars); Contrastive Search was adopted into HuggingFace `transformers` (v4.24.0) as an official decoding method and is now used by practitioners worldwide, with the associated paper (NeurIPS 2022, Spotlight) cited 460+ times.

2. LLM critique-ability evaluation and training — Shanghai AI Laboratory *Jul. 2023 – Jan. 2025 (19 months)*
Task: Establish a rigorous methodology to measure and improve large language models' ability to critique their own or others' outputs. **My role:** Principal investigator; designed the CriticEval benchmark (4 dimensions × 9 scenarios) and a multi-agent feedback framework for critique training. **Why significant:** CriticEval (NeurIPS 2024) became a reference benchmark for LLM self-critique research and was applied in the post-training of the open-sourced InternLM model family; the follow-up training method was published at EMNLP 2025 (Findings).

3. Expert-level agent benchmarking and DeepResearch agents — Alibaba Group *Aug. 2025 – present (12+ months)*
Task: Evaluate and improve LLM agents' ability to apply complex, expert-written hierarchical rules (e.g., customs/tariff classification) and to perform long-horizon information seeking. **My role:** Project lead; directed a panel of 26 domain experts to construct HSCodeComp and benchmarked 23 LLMs/agents; designed the Table-as-Search agentic information-seeking framework; contributed to Alibaba's Marco DeepResearch agent stack. **Why significant:** HSCodeComp received the **ACL 2026 Best Resource Award** (top resource paper among 12,148 submissions); both works were released as open-source benchmarks/frameworks adopted internally for agent evaluation at Alibaba.

Bibliography (Selected, 1,700+ citations on Google Scholar)

* *indicates equal contribution.*

Tian Lan*, Yiqian Yang*, Qianghuai Jia, Li Zhu, Hui Jiang, Hang Zhu, Weihua Luo, Longyue Wang. *HSCodeComp: A Realistic and Expert-level Agent Benchmark for Hierarchical Rule Application.* **ACL 2026**. **Best Resource Award.**

Tian Lan, Felix Henry, Bin Zhu, Qianghuai Jia, Junyang Ren, Qihang Pu, Haijun Li, Longyue Wang, Zhao Xu, Weihua Luo. *Table-as-Search: Agentic Information Seeking is Table Completion.* **ACL 2026, Findings**.

Yongshi Ye, Hui Jiang, Feihu Jiang, **Tian Lan**, Yichao Du, Biao Fu, Xiaodong Shi, Qianghuai Jia, Longyue Wang, Weihua Luo. *UMEM: Unified Memory Extraction and Management Framework for Generalizable Memory.* **ICML 2026**.

Rong-Cheng Tu*, Zi-Ao Ma*, **Tian Lan***, Yuehao Zhao*, Heyan Huang, Xian-Ling Mao. *Automatic Evaluation for Text-to-image Generation: Task-decomposed Framework, Distilled Training, and Meta-evaluation Benchmark.* **ACL 2025**.

Dongyang Ma, Yan Wang, **Tian Lan**. *Block-Attention for Efficient Prefilling.* **ICLR 2025**.

Chen Xu, **Tian Lan**, Yu Ji, Changlong Yu, Wei Wang, Jun Gao, Qunxi Dong, Kun Qian, Piji Li, Wei Bi, Bin Hu. *Decider: A Dual-System Rule-Controllable Decoding Framework for Language Generation.* **IEEE Transactions on Knowledge and Data Engineering (TKDE), 2025**.

Tian Lan*, Wenwei Zhang*, Chengqi Lyu, Shuaibin Li, Chen Xu, Heyan Huang, Dahua Lin, Xian-Ling Mao, Kai Chen. *Training Language Models to Critique With Multi-agent Feedback.* **EMNLP 2025, Findings**.

Tian Lan*, Wenwei Zhang*, Chen Xu, Heyan Huang, Dahua Lin, Kai Chen, Xian-Ling Mao. *CriticEval: Evaluating Large-scale Language Model as Critic.* **NeurIPS 2024**.

Tian-Yi Che, Xian-Ling Mao, **Tian Lan**, Heyan Huang. *A Hierarchical Context Augmentation Method to Improve Retrieval-Augmented LLMs on Scientific Papers.* **SIGKDD 2024**.

Tian Lan, Deng Cai, Yan Wang, Yixuan Su, Heyan Huang, Xian-Ling Mao. *Exploring Dense Retrieval for Dialogue Response Selection.* **ACM Transactions on Information Systems (TOIS), 2024**.

Tian Lan*, Deng Cai*, Yan Wang, Heyan Huang, Xian-Ling Mao. *Copy is All You Need.* **ICLR 2023**.

Yixuan Su, **Tian Lan**, Yan Wang, Dani Yogatama, Lingpeng Kong, Nigel Collier. *A Contrastive Framework for Neural Text Generation.* **NeurIPS 2022, Spotlight**.

Tian Lan, Xian-Ling Mao, Wei Wei, Xiaoyan Gao, Heyan Huang. *PONE: A Novel Automatic Evaluation Metric for Open-Domain Generative Dialogue Systems.* **ACM TOIS 2020**.

Full publication list (20+ peer-reviewed papers at ACL, EMNLP, NeurIPS, ICLR, ICML, SIGKDD, IEEE TKDE, IEEE TIP, ACM TOIS, etc.) available at Google Scholar.

Honors & Awards

- ACL 2026 Best Resource Award — for HSCodeComp (top resource paper, 12,148 submissions).
- Alistar — Alibaba Group's flagship campus recruitment program, 2025.
- National Scholarship — Beijing Institute of Technology, 2024 (Ph.D.).
- Tencent Qingyun Program — Tencent, 2025.
- Tencent Rhino-Bird Elite Intern Program — Tencent, 2022.
- National Scholarship — Beijing Institute of Technology, 2017 (undergraduate).

Contributions to the Profession

- Peer reviewer for ICLR, AAAI, NeurIPS, ACL Rolling Review (ARR), ICML, and ACM TOIS.
- Open-source contributions with sustained community adoption: Contrastive Search (merged into HuggingFace `transformers`), Copy-is-All-You-Need, CriticEval, PandaGPT, InternLM2, HammerLLM, Mozi-LLM.
- Invited talks: ACL 2026 (Alibaba Cloud, Jul. 2026); China Generative AI Summit 2026 (Apr. 2026); 1st CCF Conference on LLMs & AI Engineering (Apr. 2025).